

Detecting Heart Abnormality using ECG with CART

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1. Problem Statement

The objective of this project is to develop a machine learning model using ECG-related features to detect and classify heart abnormalities. CART (Classification and Regression Tree) is used as the primary classification approach, with Random Forest included as a comparison model. The work focuses on data preprocessing, missing-value handling, decision-tree hyperparameter tuning, model evaluation, and feature-importance analysis.

2. Dataset

The project uses the UCI Arrhythmia Dataset. The dataset contains 452 samples and 279 original attributes, with missing values represented by '?'. The target represents arrhythmia classes and the data are highly imbalanced. In the processed data used for this implementation, 14 target classes are observed. One specified feature (Feature 14 / zero-based column 13 in the implementation) was removed before model training, leaving 278 input features.

3. Methodology

The workflow begins with loading the dataset and replacing '?' with missing values. The specified feature is removed and the target column is separated from the input features. The data are divided using an 80:20 train-test split with `random_state = 42` and stratification. Median imputation is fitted on the training data and then applied to the test data. CART uses Gini impurity and is hyperparameter-tuned using cross-validation. A Random Forest with 100 estimators and `random_state = 42` is used as the comparison model.

4. Preprocessing

Preprocessing is necessary because the Arrhythmia dataset contains missing entries and mixed data characteristics. Missing entries are converted to NaN and numerical missing values are replaced using median imputation. The train-test split is performed before imputation so that the imputation statistics are learned only from training data, avoiding information leakage. The same split and preprocessing strategy are used for the Random Forest comparison.

5. Implementation

The primary CART model was tuned using Gini impurity, `max_depth` values of 3, 5, 7 and 10, `min_samples_split` values of 2, 5 and 10, and `min_samples_leaf` values of 1, 2 and 5. The final tuned configuration reported in the project notebook used Gini, `max_depth = 7`, `min_samples_split = 2`, `min_samples_leaf = 5`, together with `SelectKBest (k = 30)`. Its best cross-validation accuracy was 72.29%. Random Forest was implemented with 100 trees.

6. Results

Model	Accuracy	Precision	Recall	F1
Tuned CART	72.53%	66.64%	72.53%	68.64%
Random Forest	73.63%	67.53%	73.63%	68.49%

The tuned CART model achieved 72.53% test accuracy. The Random Forest comparison achieved 73.63% accuracy, 67.53% weighted precision, 73.63% weighted recall, and 68.49% weighted F1. Random Forest therefore produced a 1.10 percentage-point improvement in accuracy over tuned CART on this test split. CART, however, produced a marginally higher F1 score (68.64% versus 68.49%). The difference is small, so the models show broadly comparable performance on the selected test set.

7. Discussion

The results demonstrate the effect of model choice on ECG-based multi-class classification. CART is useful when interpretability is important because its decisions can be represented through feature splits and branches. Hyperparameter tuning controls tree complexity and helps avoid an unnecessarily deep tree. Random Forest combines multiple decision trees and achieved slightly stronger accuracy, precision, and recall in this experiment. The dataset's severe class imbalance should be considered when interpreting the metrics, and weighted averages were used for precision, recall, and F1.

Feature-importance analysis from CART identified Feature 14 as the highest-importance feature (importance approximately 0.2288), followed by Features 196, 90, 111 and 210. This provides an interpretable summary of which input dimensions contributed most strongly to the trained tree. Feature importance should be interpreted as model-specific rather than as a clinical conclusion.

8. Conclusion

CART was successfully implemented for ECG-based heart abnormality classification, including data cleaning, median imputation, hyperparameter tuning, evaluation, and feature-importance analysis. Random Forest provided slightly higher test accuracy than the tuned CART model, while CART achieved a marginally higher F1 score. The project therefore demonstrates both a relatively interpretable tree-based classifier and a stronger ensemble baseline on the same experimental split.

9. References

[1] UCI Machine Learning Repository, *Arrhythmia Dataset*. University of California, Irvine.

[2] Ben Zhou, Gaspar Garcia, and Paurakh Rajbhandary, *Detecting Heart Abnormality using ECG with CART*.